

Report
Of
Research & Development in Ready-Mix Concrete (RMC) Production

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

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Batch: 2024-2028

CANDIDATE'S DECLARATION

I, HARSHIT PARMAR do hereby declare that the internship report titled **“Research & Development in Ready-Mix Concrete (RMC) Production”** has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: HARSHIT PARMAR

Roll No. : 2410031422

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank Sunshine Readymix and the Research & Development (R&D) Department of their Ready-Mix Concrete Plant for providing me the opportunity and the practical exposure to complete this internship, and for the guidance extended to me throughout the three-month period.

I express my gratitude to my parents for their blessings.

Signature

Student Name : HARSHIT PARMAR

Roll No. : 2410031422

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3. INTERNSHIP COMPLETION CERTIFICATE

GSTIN: 06CNC PG9734Q2ZD



Rithoj Road, Village- Bhondsi, Haryana-122102 (Mob.+91 9599999308)

01th Sep 2026

INTERNSHIP CERTIFICATE

TO WHOMSOEVER IT MAY CONCERN

This is to certify that Mr. Harshit Parmar has successfully completed a three-month internship in the Research & Development (R&D) Department at our Ready-Mix Concrete (RMC) Plant from 1 June 2026 to 31 August 2026.

During his internship, he was involved in activities related to concrete technology, mix design, quality control, material testing, production processes, and research & development of concrete mixes. He gained practical exposure to RMC plant operations and developed an understanding of the various testing and quality parameters associated with concrete production.

His conduct and performance during the internship were found to be satisfactory. He demonstrated sincerity, enthusiasm, and a willingness to learn throughout the internship period.

We wish Mr. Harshit Parmar all the very best for his future academic and professional endeavors.

Internship Duration: 1 June 2026 – 31 August 2026

Department: Research & Development (R&D)

Industry: Ready-Mix Concrete (RMC)

For Sunshine Readymix



Authorized Signatory

4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the industrial internship undertaken in the Research & Development (R&D) Department of Sunshine Readymix, a Ready-Mix Concrete (RMC) production facility. The internship was carried out over the period of 1 June 2026 to 31 August 2026. The primary aim of this programme was to provide hands-on experience with industrial concrete production practices, specifically focusing on concrete technology, mix design, and quality control. By working within a live plant environment, the internship bridged the gap between theoretical material science concepts and practical, industry-standard implementation.

4.2 Organization Profile

The internship was hosted by Sunshine Readymix, an organization operating a Ready-Mix Concrete plant at Rithoj Road, Village Bhondsi, Haryana. The company produces factory-batched concrete of controlled quality and supplies it to construction projects in the surrounding region. Its Research & Development department is responsible for developing and validating concrete mix designs, monitoring the quality of incoming raw materials, and maintaining consistency across production batches.

4.3 Problem Statement

Ready-Mix Concrete must reach a construction site with a specified strength, workability, and durability, despite continuous variability in raw materials, ambient conditions, and transit time. The core challenge addressed during this internship was to understand how a production plant achieves consistent, specification-compliant concrete through systematic mix design, incoming material testing, controlled batching, and continuous verification of both fresh and hardened concrete properties.

4.4 Project Objectives

The core objectives of this internship were to gain practical competence across five key technical domains:

- **Concrete Technology:** To develop a working understanding of the constituent materials of concrete — cement, coarse and fine aggregates, water, and chemical admixtures — and the way each influences the properties of the final product.

- **Mix Design:** To study the proportioning of concrete mixes for target grades in accordance with established guidelines, and to understand how trial mixes are adjusted to meet strength and workability requirements.
- **Quality Control:** To observe and support the quality assurance procedures applied at each stage of production, ensuring that every batch conforms to the specified requirements before dispatch.
- **Material Testing:** To gain practical exposure to standard laboratory tests carried out on raw materials, on fresh concrete, and on hardened concrete specimens.
- **Production Processes:** To understand the end-to-end operation of an RMC plant, from raw material storage and batching through mixing, transportation, and delivery to site.

4.5 Scope of the Project

The scope of this internship was focused entirely on the production and quality assurance side of ready-mix concrete. It encompassed the complete lifecycle of a concrete batch, starting from raw material receipt and testing, through mix proportioning and computerised batching, to fresh concrete testing at the point of discharge and compressive strength verification of hardened specimens. It culminated in an understanding of how quality control records are maintained to make each batch traceable. Structural design of concrete members, site placement and compaction practices, formwork, and commercial or logistics planning were deliberately kept outside the scope of this internship to maintain a strict focus on core concrete production and quality control.

4.6 Technologies and Tools Used

Based on the practical work carried out during the internship, the following materials, methods, and equipment were central to the project:

- **Raw Materials and Constituents:** Ordinary Portland Cement and Portland Pozzolana Cement, coarse aggregate, fine aggregate, potable water, and chemical admixtures used to modify workability and setting behaviour.
- **Mix Design Methodology:** Standard concrete mix proportioning guidelines were used to establish the target mean strength for a given grade and to arrive at the corresponding water-cement ratio and material quantities.
- **Laboratory Testing Equipment:** Compression testing machine, slump cone apparatus, sieve shaker for grading analysis, weighing balances, moulds for casting specimens, and a curing tank for controlled curing.

- **Plant Systems:** The computerised batching plant control system, which governs the weighing and dosing of each constituent, together with the mixing unit and transit mixers used for delivery.

4.7 System Architecture

The plant operated as a multi-stage system, with each stage handling a distinct part of the production sequence and feeding into the next:

- **Raw Material Storage and Handling Layer:** Cement silos, segregated aggregate bins, water storage, and admixture dosing units held each constituent under controlled conditions until required for batching.
- **Batching and Weighing Layer:** The computerised control system acted as the sole entry point for a production batch, weighing every constituent to the proportions specified by the approved mix design and validating them before release.
- **Mixing and Transport Layer:** The mixing unit combined the weighed constituents into a homogeneous mix, which was then discharged into transit mixers that maintained the concrete in a workable state during delivery.
- **Quality Control and Testing Layer:** The plant laboratory verified the incoming raw materials, tested the fresh concrete at the point of discharge, and cast and cured specimens for later strength verification, providing feedback to the batching stage.

4.8 Methodology

The internship followed a sequential, progressive methodology over the course of June 2026 to August 2026.

1. **Phase 1 (Orientation and Material Study):** The internship began with an orientation to the plant layout and safety procedures, followed by a study of the constituent materials of concrete and the tests carried out on incoming consignments.
2. **Phase 2 (Mix Design and Laboratory Testing):** The focus then moved to the laboratory, where mix proportioning for target grades was studied, trial mixes were observed, and standard tests were carried out on fresh and hardened concrete specimens.
3. **Phase 3 (Production and Quality Control):** Finally, the laboratory work was tied to live production through observation of the batching process, testing of fresh concrete at the

point of discharge, verification of specimen strength at the prescribed ages, and maintenance of the associated quality control records.

4.9 Expected Outcomes

By the end of the internship, several practical and technical outcomes were successfully achieved:

- A working understanding of the constituent materials of concrete and the influence each has on strength, workability, and durability.
- The ability to follow the mix design procedure for a target grade and to appreciate how trial mixes are adjusted to meet the specified requirements.
- Practical competence in the standard tests applied to fresh and hardened concrete, including workability measurement and compressive strength verification.
- A complete understanding of the RMC plant production process, from raw material storage and batching through mixing, transport, and delivery.
- An appreciation of the quality control documentation that makes each production batch traceable and auditable.

4.10 Certificates of Completion and Communication Proof



Rithoj Road, Village- Bhondsi, Haryana-122102 (Mob.+91 9599999308)

01st Sep 2026

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Internship Duration: 1 June 2026 – 31 August 2026

Department: Research & Development (R&D)

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For Sunshine Readymix



Authorized Signatory

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1. Sunshine Readymix, “Internship Certificate,” Research & Development (R&D) Department, Ready-Mix Concrete (RMC) Plant, Rithoj Road, Village Bhondsi, Haryana – 122102, September 1, 2026.
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